

Ch. 12 Maximum Entropy

Goal: Find density ftn f^* st. ~~maximizes~~ $h(f)$ under given constraints.

Maximum Entropy
Estimation spectrum ftn Maximum Entropy rate.

12.1 Finding Maximum Entropy

$(f(x))$

$$\left[\begin{array}{l} f(x) \geq 0 \quad \& \quad S := \text{supp } f \quad f(x) = 0 \Leftrightarrow x \notin S \\ \int_S f(x) dx = 1 \\ \int_S f(x) r_n(x) dx = \alpha_n \quad n = 1, \dots, m \end{array} \right] \Rightarrow (*)$$

App 1 f^*

$$\underline{J(f)} = - \int f \ln f + \lambda_0 \int f + \sum_{n=1}^m \lambda_n \int f r_n \quad f: \text{maximum ent.}$$

$$\frac{\partial J}{\partial f(x)} = 0 = -\ln f(x) - 1 + \lambda_0 + \sum_{n=1}^m \lambda_n r_n \Rightarrow \underline{f^*(x) = e^{\lambda_0 + \sum \lambda_n r_n(x)}}$$

App 2 $\rightarrow$ KL Divergence

$$\underline{f^*(x) = e^{\lambda_0 + \sum \lambda_n r_n(x)}}$$

Thm 12.1.1 $\underline{f^*(x) = e^{\lambda_0 + \sum \lambda_n r_n(x)}}$ $\lambda_0, \dots, \lambda_m$ are chosen to satisfy $(*)$

$\Rightarrow \underline{f^*(x)}$ uniquely maximizes entropy

pf) g : probability density ftn satisfies $(*)$

$$\Rightarrow \left[\begin{array}{l} \int_S g r_n = \alpha_n = \int_S f^* r_n \\ \int_S g = 1 = \int_S f^* \end{array} \right] \Rightarrow h(g) \leq h(f^*)$$

$$\underline{h(g)} = - \int_S g \ln g = - \int_S g \ln \frac{g}{f^*} - \int_S g \ln f^* \quad \left[\begin{array}{l} D_{KL}(g \| f^*) = 0 \\ g = f^* \end{array} \right]$$

$$\leq - \int_S g \ln f^* = - \int_S g \left[\lambda_0 + \sum_{n=1}^m \lambda_n r_n(x) \right] dx \\ = - \int_S f^* \left[\lambda_0 + \sum_{n=1}^m \lambda_n r_n(x) \right] dx = \underline{h(f)}$$

$$\text{Ex 1)} \quad S = [a, b] \Rightarrow f^*(x) = \frac{1}{b-a}$$

$$\underline{f^*(x)} = e^{\lambda_0} \Rightarrow \int_S \underline{f^*(x)} dx = 1 \Rightarrow \underline{f^*(x)} = \frac{1}{b-a}$$

$$\text{Ex 2)} \quad S = \mathbb{R} \quad \underline{\mathbb{E}(x) = d_1}, \quad \underline{\mathbb{E}(x^2) = d_2}$$

$$\Rightarrow \underline{f^*(x) = \exp[\text{2nd-deg-poly}]} \Rightarrow x \sim \underline{N(d_1, d_2 - d_1^2)}$$

$$\text{Ex 3)} \quad S = \mathbb{R}^n, \quad \underline{\mathbb{E}(x_{\tilde{\alpha}} x_{\tilde{\beta}}) = K_{\tilde{\alpha}\tilde{\beta}}} \quad 1 \leq \tilde{\alpha}, \tilde{\beta} \leq n$$

$$\underline{\tilde{x}} = (x_1, \dots, x_n)$$

$$\underline{f^*(\tilde{x})} = e^{\lambda_0 + \underbrace{\sum \lambda_{\tilde{\alpha}\tilde{\beta}} x_{\tilde{\alpha}} x_{\tilde{\beta}}}_{\text{quadratic}}} \Rightarrow \underline{x \sim N(0, K)} \quad K = (K_{\tilde{\alpha}\tilde{\beta}}) \quad \Bigg\}$$

$$\Rightarrow h^*(N(0, K)) = \frac{1}{2} \log((2\pi e)^n |K|)$$

12.4 ~ 12.6 Spectrum Estimation

$\{X_t\}$: Zero mean, stationary stochastic process

Def] $R(k) = E(X_t X_{t-k})$: Invariant from t
 $\Rightarrow = E(X_{t-k} X_t) = R(-k)$ R : even

Def] $K_{n-1} = \begin{pmatrix} (R(\hat{n}-j))_{\hat{n}-j=0, \dots, n-1} \end{pmatrix} = \begin{pmatrix} R(0) & R(1) & \dots & R(n-1) \\ R(1) & R(0) & \dots & R(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ R(n-1) & \dots & R(1) & R(0) \end{pmatrix}$

Def] Spectrum density ftn $S(\lambda)$

$$\underline{S(\lambda)} = \sum_{m=-\infty}^{\infty} R(m) e^{-im\lambda} \quad : \text{indicator of process}$$

$$\begin{matrix} \nearrow \\ \downarrow \end{matrix} \quad (x_1, \dots, x_n) \rightarrow \underline{\{X_t\}}$$

Goal: Estimate $S(\lambda)$

$$\text{App 1: } \underline{S(\lambda)} = \sum_{m=-\infty}^{\infty} \underline{R(m)} e^{-im\lambda} \quad \Rightarrow \quad \underline{\hat{R}(k)} = \frac{1}{n-k} \sum_{j=1}^{n-k} x_j x_{j+k} \quad (x_1, \dots, x_n)$$

$$\boxed{R(k)=0} \text{ for large } k \quad \Rightarrow \quad n \rightarrow \infty \quad \underline{\hat{R}(k)} \Rightarrow \underline{\hat{S}(\lambda)} \rightarrow S(\lambda)$$

App 2: Burg's Attempt : $\underline{R(k)}$ for large k ,
 use $\underline{R(k)}$ which makes fewest assumptions
 $\Leftrightarrow$ Maximizes Entropy rate

Def] Differential Ent. rate,

$$h(X) = \lim_{n \rightarrow \infty} \frac{h(X_1, \dots, X_n)}{n} = \lim_{n \rightarrow \infty} h(X_n | X_1, \dots, X_{n-1})$$

Thm 12.6.1 Burg's maximum entropy rate Thm

Constraints : $\mathbb{E}(X_t X_{t+k}) = \alpha_k$ for each $k=0, \dots, p$, $\forall t$
 p -th order Gauss - Markov Process, $AR(p)$

$$\{X_t\} \text{ s.t. } X_t = - \sum_{k=1}^p a_k X_{t-k} + Z_t, \quad Z_t \sim \text{iid } N(0, \sigma^2)$$

pf) Y_t : Any stochastic process with $\mathbb{E}(X_t X_{t+k}) = \alpha_k$ for $k=0, \dots, p$, $\forall t$

X'_t : Gaussian process $\text{Cov}(X'_t, X'_{t+k}) = \text{Cov}(Y_t, Y_{t+k})$

$$h(Y_1, \dots, Y_n) \leq h(\underline{X'_1, \dots, X'_n})$$

$$= h(X'_1, \dots, X'_p) + \sum_{j=p+1}^n h(X'_j | X'_{j-1}, \dots, X'_1)$$

$$\leq h(X'_1, \dots, X'_p) + \sum_{j=p+1}^n \underline{h(X'_j | X'_{j-1}, \dots, X'_{j-p})}$$

X'_t : p -th order G-M process with $R(k) = \alpha_k$ for $k=0, \dots, p$

$$(X_t, \dots, X_{t+p-1}) \stackrel{d}{=} (X'_t, \dots, X'_{t+p-1})$$

$$h(X'_1, \dots, X'_p) = h(X_1, \dots, X_p)$$

$$h(X'_j | X'_{j-1}, \dots, X'_{j-p}) = h(X_j | X_{j-1}, \dots, X_{j-p}) = h(X_j | X_{j-1}, \dots, X_1)$$

$$\begin{aligned} h(Y_1, \dots, Y_n) &\leq h(X_1, \dots, X_p) + \sum_{j=p+1}^n h(X_j | X_{j-1}, \dots, X_1) \\ &= h(X_1, \dots, X_n) \end{aligned}$$

$\frac{1}{n} h(Y_1, \dots, Y_n) \leq \frac{1}{n} h(X_1, \dots, X_n) \Rightarrow \{X_t\}$ gives maximum entropy rate

$$h^* = \lim_{n \rightarrow \infty} h(X_n | X_{n-1}, \dots, X_1) = \lim_{n \rightarrow \infty} h(X_n | X_{n-1}, \dots, X_{n-p})$$

$$X_n = - \sum_{j=1}^p a_j X_{n-j} + \underline{z}_n \quad \Rightarrow \quad X_n | X_{n-1}, \dots, X_{n-p} \sim N(, \sigma^2)$$

$$h^* = \underline{\underline{\frac{1}{2} \log(2\pi e \sigma^2)}}$$

$$\underline{\underline{X_t}} \quad X_t = - \sum_{k=1}^p a_k X_{t-k} + z_t$$

$$\Rightarrow R(l) = \mathbb{E}(\underline{X_t} X_{t-l}) = - \sum_{k=1}^p a_k R(l-k) \quad \underline{l=1, \dots, p}$$

$$\Rightarrow - \underbrace{\begin{pmatrix} R(0) & R(1) & \dots & R(p-1) \\ R(1) & R(0) & & \\ \vdots & & \ddots & \\ R(p-1) & & & R(0) \end{pmatrix}}_{\hookrightarrow K_{p-1}} \underbrace{\begin{pmatrix} a_1 \\ \vdots \\ a_p \end{pmatrix}}_{\substack{R(-k) = R(k)}} = \begin{pmatrix} R(1) \\ \vdots \\ R(p) \end{pmatrix} \quad \text{Yule - Walker}$$

$$R(0) = \mathbb{E}(\underline{X_t} X_t) = \mathbb{E}\left(- \sum_{k=1}^p a_k X_{t-k} X_t + \underline{z_t} X_t \right)$$

$$R(0) = - \sum_{k=1}^p a_k R(k) + \underline{\underline{\sigma^2}}$$

$$\underline{\underline{S(\lambda)}}$$

$$\{z_t\} \quad z_t \sim \text{i.i.d. } N(0, \sigma^2)$$

$$\Rightarrow S_z(\lambda) = \sigma^2 \quad (= R_z(0))$$

$$\underline{S(\lambda)} = \sum_{m=-\infty}^{\infty} R(m) e^{-im\lambda} \Rightarrow \underline{R(m)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{im\lambda} S(\lambda) d\lambda$$

$$R_z(m) = \mathbb{E}(z_t z_{t-m}) \quad X_t = -\sum_{k=1}^p a_k X_{t-k} + z_t \Leftrightarrow \underline{z_t} = \underline{X_t} + \sum_{k=1}^p a_k X_{t-k}$$

$$= \mathbb{E}\left(\left(X_t + \sum_{k=1}^p a_k X_{t-k}\right) \left(X_{t-m} + \sum_{k=1}^p a_k X_{t-m-k}\right)\right)$$

$$R(n) = \mathbb{E}(X_t X_{t-n}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\lambda} S(\lambda) d\lambda \quad \checkmark$$

$$\underline{R_z(m)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(e^{-it\lambda} + \sum_{k=1}^p a_k e^{-i(t-k)\lambda} \right) \left(e^{-i(-t-m)\lambda} + \sum_{k=1}^p a_k e^{-i(-t-k-m)\lambda} \right) S(\lambda) d\lambda$$

$$\parallel = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\lambda} \left[\left| 1 + \sum_{k=1}^p a_k e^{-ik\lambda} \right|^2 \underline{S(\lambda)} \right] d\lambda$$

$$\underline{R_z(m)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\lambda} \underline{S_z(\lambda)} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\lambda} \underline{\sigma^2} d\lambda$$

$$\Rightarrow \underline{\sigma^2} = \left| 1 + \sum_{k=1}^p a_k e^{-ik\lambda} \right|^2 S(\lambda)$$

$$\Rightarrow \underline{S(\lambda)} = \frac{\sigma^2}{\left| 1 + \sum_{k=1}^p a_k e^{-ik\lambda} \right|^2}$$

$$h^* = h(X_{p+1} | X_p \dots X_1) = h(X_1, \dots, X_{p+1}) - h(X_1, \dots, X_p)$$

$$\frac{1}{2} \log((2\pi e)^{p+1} |K_{p+1}|) \quad \frac{1}{2} \log((2\pi e)^p |K_p|)$$

$$= \frac{1}{2} \log\left((2\pi e) \frac{|K_{p+1}|}{|K_p|}\right)$$

$$\underline{\hat{R}(0)} \dots \underline{\hat{R}(p)} \quad , \quad p \ll n$$